Answer

L1 : MORE ABOUT FACTORIZATION (A)

Overview:

1 3 5 8

2 4 6 7

Which one(s) are the factors of 8?

8 = 1 x 8 = (1)(8)
8 = 2 x 4 = (2)(4)

Factorize = Find the factors

∴答案格式一定係 (   )(   )

Checkpoint 1

$$a^2 + 2ab + b^2 = (a+b)^2$$

Step 1:

$$4a^2 + 16a + 16$$

$$= (2)^2(a)^2 + 2(8a) + (4)^2$$

Step 2:

$$= (2a)^2 + 2(2a)(4) + (4)^2$$

$$= [(2a) + 4]^2$$

$$= (2a+4)^2$$

Checkpoint 2

$$a^2 - b^2 = (a+b)(a-b)$$

Step 1:

$$16a^2 - 9b^2$$

$$= 4^2a^2 - 3^2b^2$$

$$= (4a)^2 - (3b)^2$$

Step 2:

$$= [(\quad) + (\quad)]\,[(\quad) - (\quad)]$$

$$= [(4a) + (3b)][(4a) - (3b)]$$

$$= (4a+3b)(4a-3b)$$

Unit Education

Let's get started!

Factorize the following expressions.

1. $a^2 + 4a + 4$

$\boxed{(a+2)^2}$

2. $4c^2 - 12c + 9$

$\boxed{(2c-3)^2}$

3. $25d^2 + 20d + 4$

$\boxed{(5d+2)^2}$

4. $9b^2 - 6b + 1$

$\boxed{(3b-1)^2}$

5. $16m^2 + 8mn + n^2$

$\boxed{(4m+n)^2}$

6. $p^2 - 10pq + 25q^2$

$\boxed{(p-5q)^2}$

7. $16a^2 - 1$

$$\begin{aligned} 16a^2 - 1 &= (4a)^2 - 1^2 \\ &= \underline{\underline{(4a+1)(4a-1)}} \end{aligned}$$

8. $36m^2 - 25n^2$

$\boxed{(6m+5n)(6m-5n)}$

9. $4a^4 - b^2c^2$

$\boxed{(2a^2+bc)(2a^2-bc)}$

10. $32x^2 - 50y^2$

$$\begin{aligned} 32x^2 - 50y^2 &= 2(16x^2 - 25y^2) \\ &= 2[(4x)^2 - (5y)^2] \\ &= \underline{\underline{2(4x+5y)(4x-5y)}} \end{aligned}$$

11. (a) Factorize $16x^2 - 8x + 1$.
(b) Hence, factorize $16y^4 - 8y^2 + 1$.

(a) $$\begin{aligned} 16x^2 - 8x + 1 &= (4x)^2 - 2(4x)(1) + 1^2 \\ &= \underline{\underline{(4x-1)^2}} \end{aligned}$$

(b) $$\begin{aligned} 16y^4 - 8y^2 + 1 &= 16(y^2)^2 - 8(y^2) + 1 \\ &= (4y^2 - 1)^2 \qquad \text{(from (a))} \\ &= [(2y)^2 - 1^2]^2 \\ &= [(2y+1)(2y-1)]^2 \\ &= \underline{\underline{(2y+1)^2(2y-1)^2}} \end{aligned}$$

12. (a) Factorize $5x^2 + 20x + 20$.
(b) Hence, factorize $5x^2 + 20x - 25$.

(a) $$\begin{aligned} 5x^2 + 20x + 20 &= 5(x^2 + 4x + 4) \\ &= 5[x^2 + 2(x)(2) + 2^2] \\ &= \underline{\underline{5(x+2)^2}} \end{aligned}$$

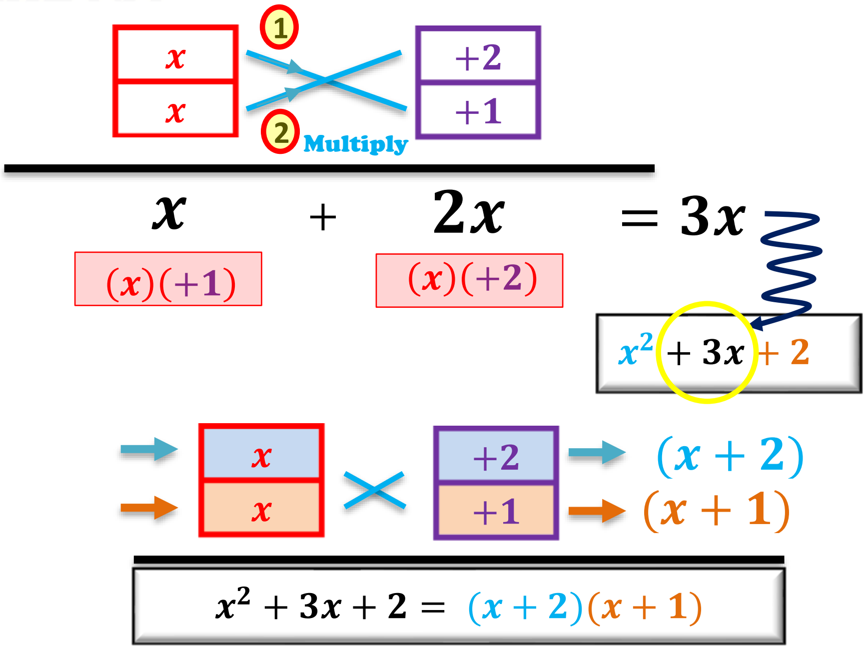
(b) $$\begin{aligned} 5x^2 + 20x - 25 &= (5x^2 + 20x + 20) - 45 \\ &= 5(x+2)^2 - 45 \qquad \text{(from (a))} \\ &= 5[(x+2)^2 - 9] \\ &= 5[(x+2)^2 - 3^2] \\ &= 5(x+2+3)(x+2-3) \\ &= \underline{\underline{5(x+5)(x-1)}} \end{aligned}$$

Checkpoint 3

Factorize $x^2 + 3x + 2$

x^2
x
x

2	
$+2$	
$+1$	



You have learnt more!

Go on!

Factorize the following expressions.

13. $x^2 + 4x + 3$

$(x+1)(x+3)$

14. $y^2 + 8y + 7$

$(y+1)(y+7)$

15. $a^2 + 6a + 5$

$(a+1)(a+5)$

16. $b^2 + 12b + 11$

$(b+1)(b+11)$

17. $c^2 + 5c + 6$

$(c+2)(c+3)$

18. $d^2 + 7d + 10$

$(d+2)(d+5)$

19. $p^2 + 8pq + 15q^2$

$(p+3q)(p+5q)$

20. $m^2 + 12mn + 35n^2$

$(m+5n)(m+7n)$

21. $a^2 + 13ab + 30b^2$

$(a+3b)(a+10b)$

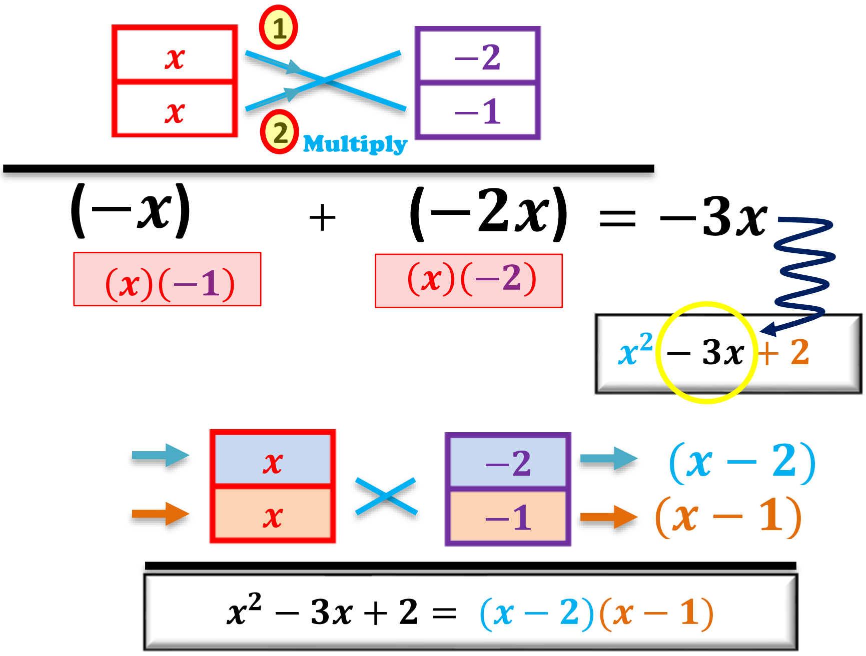
22. $x^2 + 17xy + 30y^2$

$(x+2y)(x+15y)$

Checkpoint 4

Factorize x^2 - 3x + 2

x^2	2	
x	$+2$	-2
x	$+1$	-1



Keep Fighting!

Factorize the following expressions.

23. $x^2 - 14x + 13$

$\boxed{(x-1)(x-13)}$

24. $y^2 - 13y + 12$

$\boxed{(y-1)(y-12)}$

25. $w^2 - 8w + 12$

$\boxed{(w-2)(w-6)}$

26. $z^2 - 7z + 12$

$\boxed{(z-3)(z-4)}$

27. $p^2 - 11p + 28$

$\boxed{(p-4)(p-7)}$

28. $q^2 - 16q + 28$

$\boxed{(q-2)(q-14)}$

29. $r^2 - 9rs + 20s^2$

$\boxed{(r-4s)(r-5s)}$

30. $a^2 - 18ab + 56b^2$

$\boxed{(a-4b)(a-14b)}$

31. $s^2 - 19st + 70t^2$

$\boxed{(s-5t)(s-14t)}$

32. $\alpha^2 - 16\alpha\beta + 48\beta^2$

$\boxed{(\alpha-4\beta)(\alpha-12\beta)}$

Unit Education

Factorize the following expressions.

33. $a^2 + 5a + 6$

$\boxed{(a+2)(a+3)}$

34. $a^2 + 5 + 6a$

$\boxed{(a+1)(a+5)}$

35. $b^2 - 12b + 27$

$\boxed{(b-3)(x-9)}$

36. $3b^2 - 36b + 81$

$\boxed{3(b-3)(b-9)}$

37. $x^2 - 15x + 36$

$\boxed{(x-3)(x-12)}$

38. $2x^2 + 30x + 72$

$\boxed{2(x+3)(x+12)}$

39. $m^2 - 10mk + 21k^2$

$\boxed{(m-3k)(m-7k)}$

40. $m^2n - 10mnk + 21k^2n$

$\boxed{n(m-3k)(m-7k)}$

41. $s^2 + 15st + 26t^2$

$\boxed{(s+2t)(s+13t)}$

42. $26s^2 - 15st + t^2$

$\boxed{(t-2s)(t-13s)}$

43. $5c^2 - 40cd + 75d^2$

$\boxed{5(c-3d)(c-5d)}$

44. $p^3 + 18p^2t + 45pt^2$

$\boxed{p(p+3t)(p+15t)}$

45. $u^2t^2 + 13uvt + 22v^2$

$\boxed{(ut+2v)(ut+11v)}$

46. $x^4 - 16x^2y^2 + 63y^4$

$$\boxed{\begin{aligned}&(x^2-7y^2)(x^2-9y^2)\\=\ &(x^2-7y^2)(x+3y)(x-3y)\end{aligned}}$$

Give it a try

Expand the following expressions.

$(a+1)(a+2)$　　　　$(a-1)(a-2)$　　　　$(a+1)(a-2)$

最尾係正，正正負負

最尾係負，正負正負

Checkpoint 5

Factorize x^2+2x-3

Trial 1

x^2
x
x

-3	
$+3$	-3
-1	$+1$

x	① ② Multiply	$+3$
x		-1

$(-x) + (3x) = 2x$

$(x)(-1)$　　$(x)(+3)$

$x^2 + 2x + 3$

Trial 2

x^2
x
x

-3	
$+3$	-3
-1	$+1$

x	① ② Multiply	-3
x		$+1$

$(x) + (-3x) = -2x$

$(x)(+1)$　　$(x)(-3)$

$x^2 + 2x + 3$

Final Ans:

$$x^2+2x-3=(x+3)(x-1)$$

Unit Education

Factorize the following expressions.

47. $x^2 + 2x - 3$

$\boxed{(x+3)(x-1)}$

48. $x^2 - 2x - 3$

$\boxed{(x+1)(x-3)}$

49. $y^2 + 3y - 10$

$\boxed{(y+5)(y-2)}$

50. $y^2 - 3y - 10$

$\boxed{(y+2)(y-5)}$

51. $h^2 - 14h - 51$

$\boxed{(h+3)(h-17)}$

52. $n^2 + 16n - 57$

$\boxed{(n+19)(n-3)}$

53. $r^2 - 8r - 33$

$\boxed{(r+3)(r-11)}$

54. $k^2 + 3k - 54$

$\boxed{(k+9)(k-6)}$

55. $x^2 - 11xy - 42y^2$

$\boxed{(x+3y)(x-14y)}$

56. $a^2 - 9ab - 52b^2$

$\boxed{(a+4b)(a-13b)}$

57. $c^2 + 19cd - 120d^2$

$\boxed{(c+24d)(c-5d)}$

58. $p^2 + 14pq - 72q^2$

$\boxed{(p+18q)(p-4q)}$

Final Boss!

Factorize the following expressions.

59. $c^2 - 28d^2 + 3cd$

$\boxed{(c + 7d)(c - 3d)}$

60. $p^2 + 40q^2 - 14pq$

$\boxed{(p - 4q)(p - 10q)}$

61. $-6st + s^2 - 91t^2$

$\boxed{(s + 7t)(s - 13t)}$

62. $-r^2 + 17rs - 70s^2$

$\boxed{-(r - 7s)(r - 10s)}$

63. $-u^2 - 13uv + 68v^2$

$-(u + 17v)(u - 4v)$
or $(17v + u)(4v - u)$

64. $-68n^2 - 21mn - m^2$

$\boxed{-(m + 4n)(m + 17n)}$

65. $x^2 - 36y^2 - 9xy$

$\boxed{(x + 3y)(x - 12y)}$

66. $3a^2 - 15ab - 42b^2$

$\boxed{3(a + 2b)(a - 7b)}$

67. $2p^2 - 19pq + 45q^2$

$\boxed{(2p - 9q)(p - 5q)}$

68. $3h^2 - 16hk - 35k^2$

$\boxed{(3h + 5k)(h - 7k)}$

69. $-119\theta\phi - 42\theta^2 + 98\phi^2$

$\boxed{-7(2\theta + 7\phi)(3\theta - 2\phi)}$

70. $24\alpha^4\gamma^2 + 90\beta^4\gamma^2 - 94\alpha^2\beta^2\gamma^2$

$$\begin{aligned}&2\gamma^2(12\alpha^4 - 47\alpha^2\beta^2 + 45\beta^4)\\ &= 2\gamma^2(4\alpha^2 - 9\beta^2)(3\alpha^2 - 5\beta^2)\\ &= 2\gamma^2(2\alpha + 3\beta)(2\alpha - 3\beta)(3\alpha^2 - 5\beta^2)\end{aligned}$$